# RAGHAV RAMAN

Data Engineer | ETL & Data Integration | AI Engineer | Informatica IICS, SQL, Python


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Portfolio: theraghavraman.github.io/the_raghav_raman


SCAN FOR PORTFOLIO

## PROFESSIONAL SUMMARY


Senior Software Engineer at Capgemini with 4+ years in data engineering: ETL pipelines on Informatica IICS and PowerCenter, Snowflake and SQL, turning raw data into decisions. Outside work I build and ship software, including ReData, an incremental data-ingestion engine on DuckDB. Now going deeper into AI engineering: AMD ROCm Certified Associate, DeepLearning.AI Deep Learning Specialization, and Knowledge Graphs for RAG.


## TECHNICAL SKILLS

**ETL & Data Integration:** Informatica IICS/IDMC, PowerCenter, Informatica Admin, SSIS, Talend, ETL Pipelines, SCD Type 1/2, Hierarchy Parser, Data Mapping, Data Validation, Flat-File Ingestion, Incremental Sync

**Databases & SQL:** SQL, Oracle, IBM DB2, SQL Server (SSMS), Snowflake, MySQL, DuckDB, MongoDB

**Programming & APIs:** Python, FastAPI, REST APIs, SQLAlchemy, Pydantic, Pytest, Celery, Redis, JavaScript, TypeScript

**Cloud & Big Data:** Azure Synapse, Azure VDI, AWS, GCP, BigQuery, Databricks, Apache Kafka, Apache Arrow, Docker

**AI & Retrieval:** RAG, LangChain, FAISS, Embeddings, OpenAI API, AI Agents, Prompt Engineering

**Data Reliability & Ops:** Pipeline Monitoring, ServiceNow, Incident Mgmt, ITIL, Root-Cause Analysis

**Tools & Platforms:** Git, GitHub, GitHub Actions, Linux/Unix, Nginx, Windows, macOS, Microsoft Excel, BI Reporting

## PROFESSIONAL EXPERIENCE

**Senior Software Engineer** | Capgemini — Oct 2026 – Present

- Collaborated with the team on bridge calls for Informatica updates, coordinating changes and resolving issues live.
- Supported a database migration with the team, moving and verifying ETL jobs and data on the new database.

**Analyst** | Capgemini — Oct 2022 – Oct 2026

- Built and maintained ETL pipelines in Informatica PowerCenter and Informatica Intelligent Cloud Services (IICS) for client data platforms.
- Loaded and managed data in Snowflake, and automated routine jobs with shell scripts.
- Built Informatica Cloud (IICS) mappings and tasks for SCD Type 1, SCD Type 2, file handling and Hierarchy Parser use cases on Oracle and flat-file connections.
- Used Router, Aggregator, Normalizer, Joiner, Lookup, Expression, Update Strategy and Sequence Generator transformations; validated loads with row counts and rejected-row checks.
- Worked with business and client teams to turn requirements into reliable data flows, and prepared clean, validated datasets for BI reporting and analytics.
- Reviewed use cases from technical and functional perspectives with teammates; shortlisted for a BFSI data-integration engagement.

## PROJECTS

**ReData: Incremental Data-Ingestion Engine** — Python | DuckDB | Apache Arrow | FastAPI | CLI

Repo:github.com/theraghavraman/redata| Live demo: theraghavraman.github.io/redata

- Built an engine that copies database tables, files and API responses into DuckDB and keeps them current, with connectors for PostgreSQL, MySQL, SQL Server, Snowflake, SQLite, S3-compatible buckets and JSON REST APIs, read as Arrow batches of 50,000 rows.
- Implemented three sync modes (full refresh that builds beside the old table and swaps it in, append by cursor, upsert by key in one transaction), saving cursor state after every successful run and handling schema drift by adding columns and widening types.

- Exposed it through a CLI, a FastAPI HTTP API and a scheduler with a run log queryable in SQL; `redata sync --all` exits with status 1 if any sync fails, so it can run unattended from cron.

### ETL Validation Toolkit

Python | Pydantic | Pytest | MySQL

Repo: github.com/theraghavraman/etl-validation-toolkit

- Created a reusable validation layer that cleans CSV, Excel and API feeds before MySQL loading, with Pydantic schema checks, typed models, rejected-row reporting and automated tests.

### CodeProof: Browser-First Code Verification Lab

JavaScript | TypeScript | Web Workers | Pyodide | GitHub Actions

Repo: github.com/theraghavraman/codeproof | Live site: theraghavraman.github.io/codeproof

- Built a tool that inspects, type-checks and test-runs unfamiliar JavaScript, TypeScript and Python entirely in the browser tab, so code is never uploaded to a server.
- Ran static rules, a TypeScript type check and import-graph mapping in a Web Worker, executed tests only inside a sandboxed iframe, and ran Python through a pinned Pyodide runtime; findings export as JSON and Markdown evidence reports.
- Set up CI that rebuilds the bundled TypeScript compiler from verified parts and checks every hash before deploying.

### RAG Document Q&A API

Python | FastAPI | LangChain | FAISS | Redis | Celery

- Built a document question-answering API covering ingestion, chunking, embeddings, FAISS vector indexing and grounded retrieval.
- Moved ingestion to Celery/Redis background jobs and returned structured JSON answers through a REST endpoint.

### AI Code Testing & Bug Investigation Agent

Python | FastAPI | OpenAI API | Pytest | Pydantic | Docker

- Built a tool-calling AI agent that reads source and specification files, generates targeted Pytest cases and runs them to investigate bugs.
- Sandboxed generated tests in Docker with resource and time limits and no network access; produced JSON/Markdown reports separating test evidence from model analysis.

## EDUCATION

### Bachelor of Technology (B.Tech), Electronics and Communication Engineering

Lakshmi Narain College of Technology (LNCT), Bhopal – RGPV | 2022 | CGPA 7.89/10

## CERTIFICATIONS

- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional – Oracle
- IDMC Foundation Certification; Cloud Data Integration for PowerCenter Developers (Foundation) – Informatica
- Generative AI Fundamentals Accreditation – Databricks Academy
- Google Advanced Data Analytics Professional Certificate – Google
- AWS Cloud Quest: Cloud Practitioner; Generative AI Essentials – Amazon Web Services
- Snowflake Data Warehousing Workshop – Snowflake
- Confluent Fundamentals: Apache Kafka – Confluent

## ADDITIONAL HIGHLIGHTS

- Completed about 40 hands-on labs in Google Cloud Computing Foundations and 16 labs in the Google Cloud Hero challenge; finished AWS Cloud Quest with 12 DIY assignments.
- Completed the LevelUp SkillUp business-communication program (Mar – Jul 2024).